

Analyzing Housing Affordability and Living Risks Across U.S. States

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Abstract— The modern United States housing market is influenced by an increasingly complex array of socio-economic and environmental variables. Traditional housing platforms focus heavily on median prices and income levels but frequently omit crucial livability metrics such as climate risk, local crime rates, home insurance costs, and air quality (PM2.5). This electronic document details the architecture, implementation, and findings of the "U.S. Housing Environment Analysis Dashboard." By synthesizing disparate datasets from the U.S. Census Bureau, the Environmental Protection Agency (EPA), the Federal Emergency Management Agency (FEMA), and the Federal Bureau of Investigation (FBI), we provide a holistic, interactive analytical tool. Furthermore, our system integrates a Generative Artificial Intelligence (AI) assistant to facilitate dynamic data exploration. The resulting platform enables users to perform granular risk assessments and uncovers latent correlations between economic prosperity and environmental safety.

Keywords— *artificial intelligence, climate risk, data visualization, housing market, interactive dashboards.*

I. Introduction

Housing affordability has become an increasingly critical issue across the United States, driven by rising home prices, income disparities, and environmental and social risks. While many studies focus on individual factors such as income or housing price alone, real-world affordability is influenced by a complex combination of economic, environmental, and safety-related conditions. Existing analyses often evaluate housing affordability using isolated variables, making it difficult to understand the combined impact of economic, environmental, and social risks on overall living conditions.

To address this gap, this project aims to analyze how housing prices, income, and environmental/living risks interact to affect affordability and quality of living across U.S. states. Specifically, we integrate multiple datasets—including housing prices, median income, air quality (PM2.5), climate risks, crime rates, and insurance costs—into a unified analytical dashboard. The primary goal of this project is to provide users with an interactive tool that enables them to compare housing affordability across states, understand the nuanced relationship between income and cost of living, evaluate environmental and safety risks, and systematically identify regions with the best overall living conditions.

A primary innovation of this research is the integration of Generative Artificial Intelligence (AI) to enhance data accessibility. Traditional dashboards often require users to possess a degree of data literacy to extract meaningful insights from complex visualizations. By incorporating the Google Gemini API, our system transitions from a passive reporting tool to an active analytical assistant. This feature allows users to engage with the data through natural language queries, enabling the immediate synthesis of statistical summaries and the identification of regional anomalies that might otherwise remain overlooked in a standard tabular format.

II. Related Work

Previous research on housing affordability has traditionally been anchored in macroeconomic indicators. Academic literature has extensively modeled housing market dynamics, focusing primarily on the fundamental relationship between housing prices, mortgage rates, and median household income levels [4], [5], [6]. Data from governmental sources, such as the U.S. Census Bureau, has been widely utilized to construct traditional affordability indices, such as the price-to-income ratio. While these baseline economic distributions provide a foundational

understanding of market accessibility, they operate under the assumption that financial capacity is the sole driver of housing decisions, often ignoring the complex realities of modern living conditions.

However, recent literature emphasizes that real-world affordability and property desirability are heavily impacted by environmental externalities. Environmental economic studies have demonstrated a statistically significant negative correlation between PM2.5 air pollution exposure and localized housing valuations [7]. These studies indicate that as public awareness of long-term health risks increases, homebuyers are increasingly factoring air quality into their purchasing decisions, leading to a depreciation of property values in highly polluted, yet economically active, urban centers.

Furthermore, the escalating frequency of natural disasters has introduced severe systemic vulnerabilities into the real estate market. Recent research highlights the danger of "unpriced climate risks," warning that millions of properties in the United States may be significantly overvalued because markets fail to adequately price in the threat of floods and wildfires [8]. As a direct consequence, property owners are facing drastically inflated home insurance premiums, which act as a hidden, recurring financial burden [9]. This climate-driven insurance crisis essentially functions as a secondary tax, fundamentally altering the true affordability of a home beyond its initial listing price.

On the social front, localized crime rates remain a critical variable that fundamentally depresses neighborhood housing demand and overall livability. Sociological and economic analyses consistently show that elevated crime rates not only reduce property values but also alter demographic migrations, pushing demand toward safer, albeit more expensive, regions [10]. Despite its importance, crime data is typically analyzed in isolation by criminologists rather than integrated directly into financial affordability calculators.

While these prior studies provide valuable, in-depth insights into individual socio-economic and environmental factors, they frequently lack an integrated, multidisciplinary approach. Most existing research presents static findings isolated within their specific academic silos. Our project differentiates itself by aggregating these disparate economic, environmental, and safety-related variables into a single, cohesive interactive platform. By leveraging modern data visualization frameworks and AI-assisted analytics, our dashboard allows users to evaluate housing affordability from a multidimensional

perspective, bridging the gap between theoretical research and practical data exploration.

III. Implementation Details

A. Data Sources

The data set used in this project is compiled from multiple public sources:

- U.S. Census Bureau – Median household income and housing prices
- EPA (Environmental Protection Agency) – PM2.5 air quality data
- FEMA – Climate risk and natural disaster frequency
- Crime Data Sources – State-level crime rates
- Insurance Data – Average home insurance costs

These datasets were cleaned, merged, and standardized into a single dataset (final_dataset.csv).

B. Data Preprocessing

Data preprocessing was performed using Python and Pandas. Key steps include:

To ensure data consistency and reliability, several preprocessing steps were systematically executed. These steps included renaming columns for structural uniformity (such as converting "State_Name" to "state"), rigorously casting numerical columns, and handling any type errors that arose during data ingestion. Furthermore, missing values were replaced with median values to prevent statistical skewing, and state names were standardized and mapped to their respective abbreviations.

The resulting final dataset comprises six primary variables: median home value (price), median household income (income), air pollution level (pm25), climate risk index (climate), crime rate (crime), and average home insurance cost (insurance).

C. Methodology (Tools & Techniques)

The system's architecture relies on a robust stack of modern data science tools. Python serves as the core programming language, utilizing Pandas for comprehensive data manipulation and preprocessing. For the user interface, Streamlit was selected as the interactive web application framework, seamlessly paired with Plotly to render dynamic data visualizations. Additionally, Google Gemini

API was integrated to function as an AI-powered analytical assistant.

The overarching methodology centers on harmonizing multiple public datasets, conducting data cleaning, and translating complex transformations into accessible insights through these interactive visualizations.

D. System Design

The project is deployed as an interactive Streamlit dashboard designed for fluid user navigation. The application features a dedicated sidebar equipped with core functionalities, including page navigation, a multi-select state filter for targeted regional analysis, and a dynamic insights panel.

To allow users to explore different aspects of housing affordability efficiently, the dashboard is modularly divided into seven distinct pages: an integrated Overview, alongside dedicated analytical sections for Housing Price, Income, Insurance, Climate Risk, Crime Rate, and PM2.5 Air Quality. This structured design ensures that users can intuitively navigate through multidimensional data.

E. Key Visualizations

The dashboard includes analytical visualizations:

Figure 1 displays the geographic distribution of housing prices across U.S. states. Darker regions indicate higher median home values, with significantly higher prices observed in coastal states such as California and the Northeast.

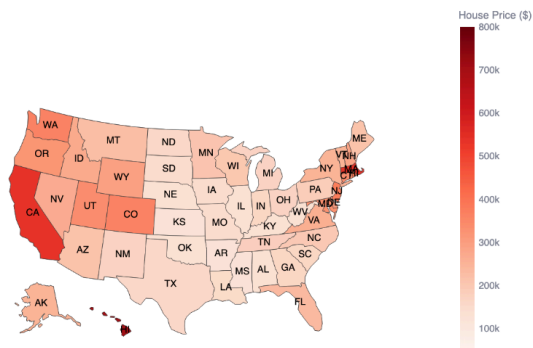


Figure 1. U.S. Housing Price Distribution

Figure 2 presents a dual-line trend chart comparing median income levels with average home insurance costs. This visualization reveals

underlying financial burden patterns, highlighting regions where insurance premiums disproportionately impact local earnings.

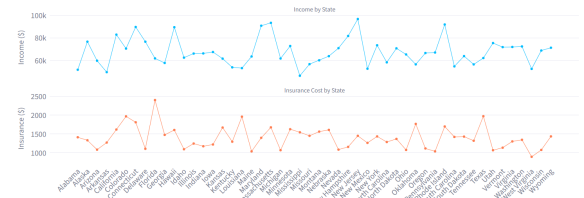


Figure 2. Income vs. Insurance Cost

Figure 3 is a histogram comparing PM2.5 levels across U.S. states, utilizing the default dataset order. Higher values indicate poorer air quality, which may impact overall living conditions and long-term health risks.

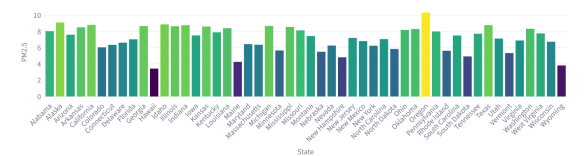


Figure 3. PM2.5 Air Quality

Figure 4 features a stacked bar chart representing an Integrated Risk Assessment. This visualization effectively combines the climate risk index and crime rate into a single, comprehensive risk metric for each state.

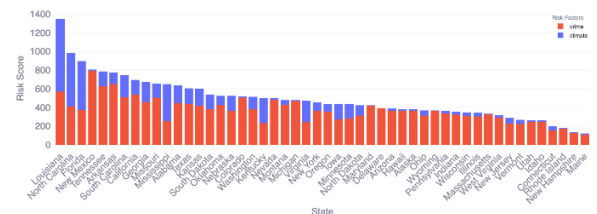


Figure 4. Integrated Risk Assessment

Figure 5 is a scatter plot with an integrated trendline that analyzes the relationship between median income and housing prices. This visualization highlights regional affordability trends, illustrating which states deviate significantly from the expected income-to-price ratio.

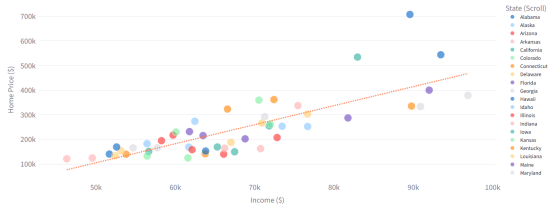


Figure 5. Price vs Income & The Affordability Trend

Figure 6 presents a correlation heatmap displaying the relationships between all key variables. This matrix allows for the quick identification of underlying socio-economic and environmental correlations, such as the potential impact of risk on home insurance costs.

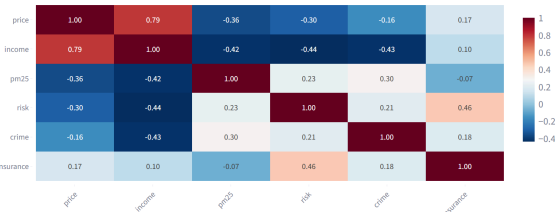


Figure 6. Variable Correlation Heatmap

All figures are generated from the interactive dashboard developed in this project.

F. Sample Inputs and Outputs

To demonstrate the system's interactive capabilities, users can provide sample inputs via the sidebar filter, such as selecting specific states (e.g., California and Texas). In response, the dashboard dynamically generates a customized suite of sample outputs. These outputs include a choropleth map detailing housing price distributions, scatter plots illustrating income versus housing price relationships, and integrated risk comparison charts combining climate and crime data. Furthermore, the system synthesizes AI-generated insights based exclusively on the selected data subsets, allowing users to interactively explore localized patterns and variables.

G. Affordability Model

To quantitatively evaluate housing affordability, we established a straightforward mathematical model. The affordability index is defined as the ratio of the median home price to the median household income:

$$Affordability = \frac{Price}{Income}$$

This metric allows for a standardized comparison of how expensive housing is relative to local income across different states. A lower calculated value indicates superior housing affordability, whereas a higher value signifies a greater financial burden on the local population.

H. AI Data Assistant

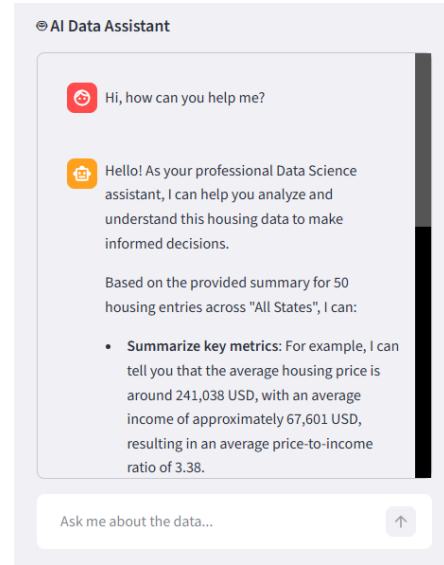


Figure 7. The Interactive AI Data Assistant Interface

A defining feature of this dashboard is the integration of an AI-powered assistant utilizing the Google Gemini API. This module allows users to pose natural language questions regarding the dataset, receive data-driven insights, and obtain recommendations based on multivariable analysis. To ensure the accuracy and relevance of its responses, the assistant dynamically ingests the current dataset summary alongside any active user-selected filters, thereby generating analytical responses.

As demonstrated in Figure 7, the chat interface provides an intuitive conversational experience. When prompted, the assistant actively acknowledges the user's current filter context (e.g., analyzing housing entries across "All States") and instantly computes key macroeconomic metrics. For instance, it can autonomously calculate the national average price-to-income ratio based on real-time data summaries.

I. Code and Data Availability

The full implementation of this project, including all source code and processed datasets, is available on GitHub: https://github.com/Cheng-Han-Chung/2026_Spring_261.

The repository contains the core Streamlit dashboard application script (app.py), data preprocessing scripts, the final integrated dataset (final_dataset.csv), and supporting notebooks utilized for data cleaning and transformation. Providing open access to these comprehensive resources ensures the full reproducibility of our analytical results and maintains strict transparency in our methodology.

J. Testing Environment and Scalability Justification

The system was developed and tested locally on a high-performance machine equipped with 64GB of SK Hynix DDR5 RAM, ensuring sufficient memory capacity for handling memory-intensive data preprocessing tasks and rendering interactive Plotly visualizations without latency. The software stack operates on Python 3.x, utilizing Streamlit's local server for continuous deployment and UI testing.

Our testing workflow consisted of three main phases: (1) Data Validation: Verifying Pandas data cleaning scripts to ensure accurate missing value imputation. (2) Component Testing: Iteratively testing Streamlit sidebar filters to confirm dynamic and accurate updates of maps and analytical charts. (3) UX Testing: Simulating interactions with the AI Assistant to ensure accurate, data-driven responses without formatting errors.

Regarding scalability, the solution is explicitly designed to handle increasing problem sizes without significant performance degradation. The data preprocessing pipeline relies heavily on Pandas' vectorized operations, which can computationally process millions of rows exponentially faster than standard iterations. On the frontend, the Streamlit dashboard employs the `@st.cache_data` decorator. This mechanism temporarily caches data loading and computational results, preventing redundant processing during complex user interactions. Consequently, as the dataset expands in the future, such as shifting to high-resolution county-level data or integrating live real-time API streams—the system architecture remains highly scalable and responsive.

IV. Results and Discussion

Through our interactive dashboard, we identified several critical insights regarding the cause-and-effect

relationships between economic capabilities, housing costs, and environmental vulnerabilities across the U.S.

The Affordability Paradox: Our analysis demonstrates that a higher median income does not inevitably equate to better housing affordability. As visualized in our scatter plots, coastal states like California and Massachusetts exhibit high median incomes but are severely offset by disproportionately higher housing prices, creating a substantial financial barrier. Conversely, several Midwestern states provide a more optimal balance, where moderate earnings are paired with significantly lower housing costs, resulting in genuinely higher affordability.

Climate Risk and the Hidden Cost of Insurance: A strong causal relationship was observed between geographic climate risks and the financial burden of home insurance. States highly susceptible to natural disasters, such as Florida and Louisiana, show a sharp, corresponding spike in average home insurance premiums. This observation aligns with recent warnings regarding unpriced climate risks in U.S. property markets [8], [9], indicating that environmental vulnerability translates directly into a hidden, recurring economic cost for homeowners that goes beyond the initial property purchasing price.

Trade-offs in Quality of Living: When integrating crime rates and PM2.5 air quality data, we observed a complex trade-off. Some regions with high housing demand and economic prosperity also face elevated living risks, such as urban density correlating with higher PM2.5 exposure or localized crime rates. This highlights that homebuyers cannot rely on price alone as a proxy for living quality; they must continuously navigate the trade-offs between financial affordability, physical safety, and environmental health.

V. Conclusion and Future Work

The core objective of this project was to untangle the multifaceted nature of housing affordability across the United States. Our multidimensional analysis reveals that higher median income does not universally guarantee better housing affordability, as demonstrated by the stark disparities in coastal regions where housing costs disproportionately outpace earnings. Furthermore, the findings underscore that environmental risks and crime rates critically impact overall living quality. Specifically, a strong correlation was observed between localized climate risk levels and escalating home insurance costs, indicating that environmental vulnerability translates directly into recurring financial burdens for homeowners that extend far beyond the initial purchase price.

Despite the inherent complexities of these socioeconomic and environmental variables, this project makes significant contributions by successfully integrating multiple heterogeneous datasets into a unified, interactive analytical framework. The developed dashboard provides a comprehensive platform that moves beyond traditional, isolated economic indicators. A critical innovation of this system is the integration of an AI-assisted exploration feature utilizing the Google Gemini API. This functionality lowers the barrier to data literacy, enabling users to interactively generate real-time insights and evaluate housing conditions from a holistic, data-driven perspective through natural language queries.

However, it is essential to acknowledge certain methodological limitations inherent in the current scope of this study. The datasets were compiled from various independent public sources which may introduce slight temporal misalignments. Additionally, the reliance on median values for missing data imputation, while statistically sound for maintaining baseline stability, might obscure localized extremes. Furthermore, aggregating data at the state level and utilizing simplified risk indicators may not fully encapsulate the hyper-local complexity of real-world housing dynamics, where neighborhood-level factors often drive market fluctuations.

Moving forward, several enhancements can substantially elevate the platform's analytical rigor and practical utility. Future iterations will aim to incorporate real-time data updates via live APIs, ensuring the dashboard reflects the most current market and environmental conditions. Transitioning towards a more granular, city-level or ZIP-code-level analysis will also provide much-needed resolution to uncover localized affordability paradoxes. Ultimately, integrating advanced machine learning models for predictive forecasting and refining risk scoring methodologies will position this project for successful future deployment as a robust, public-facing web application for both homebuyers and policymakers.

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